ReAct Agent Report

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Section 1: Implementation Logic

Prompt strategy:

Overview of the Prompt given to LLM:

- An identity which is a ReAct agent
- A set of rules/invariants it must stick to and lock desired behavior
- Provided schema / Format for reponses of the LLM
- A set of examples to manage the decomposition of problems , hard to get data and ambigous data and decision making.

Explanation:

The Model have a stochastic behavior and its output will be similar to what its training had, so by building explicit schema of the responses, the model will learn to answer in that style. Also without clear constrains, the model will likely output different responses based on arbitrary things so the set of rules and invariants are given to take away this and make the responses more deterministic. The rules are very important because they help with the decision making and ambiguity handling, this rules take away some of the agent autonomy in exchange for better behavior. For example without "If ambiguity exists → refine search", the agent will choose one of the sources from the content , even if the answer was not clear. "NEVER give Final answer before at least one Action" Avoid that the agent responds with its internal knowledge with can be outdated or hallucinated. Also a rule to choose numerical data when variation was only 10% was added so agent wont take the variation as an excuse to keep requesting data.

The examples given have ReAct pattern, so again the model learns how to decompose the problems and acquire to the format.

By passing schema of responses, Rules, examples and identity (ReAct agent), I am avoiding that the agent reason with its internal knowledge and seeks outside sources to avoid hallucination or inaccurate facts.

In the examples, I tried to simulate similar examples so the model can learn how to deal with them and with the rules, should be able to take good decisions about if time to do an action or not.

Code Snippet of initial Prompt

```
 # System prompt (ReAct + strict format)

        self.system_prompt = """

You are a ReAct agent.
You MUST follow this loop strictly:
Thought → Action → Observation → Thought → Action ...

Rules:
- NEVER skip Action after Thought
- NEVER output multiple Thoughts in a row
- NEVER give Final answer before at least one Action
- If Observation is empty or bad → REFLECT and try a better query
- If ambiguity exists → refine search
- If results have variations of 10%, choose any and compute with it (don't ask
for more data)

FORMAT STRICTLY:

Thought: <reasoning>
Action: Search["query"]

When enough information is gathered and no ambiguity:

Final: <answer>

If uncertain and ambiguity exist:

Request more data or reflect on better query. If after 5 loops still
uncertain:

Final_Uncertain: <answer>

DO NOT output anything else.
"""

        # Few-shot examples (critical)
        self.few_shot = """
User: What is the population of Germany?
```

```
Thought: The user asks for population. I should search for current data.
Action: Search["Germany current population"]

Observation: Germany population is approximately 83 million.

Thought: I have the answer.
Final: Germany's population is approximately 83 million.

---

User: What fraction of Germany's population is Spain's population?

Thought: I need both populations. I will search Germany first.
Action: Search["Germany current population"]

Observation: Germany population is 83 million.

Thought: Now I need Spain population.
Action: Search["Spain current population"]

Observation: Spain population is 49 million.

Thought: I can compute the fraction.
Final: Spain's population is about 49/83 ≈ 0.59 of Germany's population.

---

User: Japan population 2025 exact?

Thought: The query may be too specific. I will search broadly.
Action: Search["Japan population 2025"]

Observation: No results found.

Thought: The search returned nothing. I was too specific. I will broaden it.
Action: Search["Japan current population"]

Observation: Japan population is 125 million.

Thought: I now have a reasonable estimate.
Final: Japan's population is approximately 125 million.

---

User: Who is the best trader in the world?
```

```
Thought: I will search for top traders in the world.
Action: Search["Best trader in the world"]

Observation: No single answer, but some names that come up are Warren Buffett,
George Soros, and Ray Dalio.

Thought: The search returned Too many names, I will search for best trader in
2025 to narrow it down.
Action: Search["Best trader in the world 2025"]

Observation: Several answers but Warren Buffett is often cited as the best
trader in the world.

Thought: I now see warren buffett is the most consistent answer.
Final: Warren Buffett is often considered the best trader in the world,
especially in 2025.

"""
```

Loop Mechanism:

The loop works by creating implicit states for the Agent, it is forced to acknowledge ambiguity, more than one entity or lack of information and based on this states it will call for Actions after its though. The agent prompt avoids that it delivers a final answer until its certain (based on the rules and training). The internal state happens on the thought or relaxion. The state is implicit because there is not an explicit flag or variable that keeps track of it but its reflected on the messages passed to the model.

The Observations are fed back to the agent appended in the messages as a user input.

Code Snippet for Feeding Content/Observation back into the LLM context window

```
# --- TOOL EXECUTION ---
            observation = search(query)

            if not observation:
                observation = "No results found."

            print(f"Observation: {observation}")

            # --- APPEND BACK TO LLM ---
            messages.append({"role": "assistant", "content": output})
            messages.append({"role": "user", "content": f"Observation:
```

```
{observation}"})
```

Section 2: Benchmark Traces (The Evidence)

Task 1: Planning & Quantitative Reasoning

```
(age_ncu) ghostrader@Guille:~/AGE_NCU/assig2$ python main.py
ReAct Agent Started. Type 'exit' to quit.
Enter your query: What fraction of Japan's population is Taiwan's population as of 2025?

--- Iteration 1 ---
Thought: The user is asking for a fraction of populations in 2025. I need the population of Japan in 2025 and the population of Taiwan in 2025. I will start by sear
ching for Japan's population in 2025.
Action: Search["Japan population 2025"]
Observation: Title: Japan Population, 1899 – 2025 | CEIC Data
URL: https://www.ceicdata.com/en/indicator/japan/population
Content: Japan Population reached 123.2 million people in Dec 2025, compared with the previously reported figure of 123.7 million people in Dec 2024.

Title: Population of Japan 2025 - PopulationPyramid.net
URL: https://www.populationpyramid.net/japan/2025/
Content: Population pyramid of Japan in 2025. Total population: 123103479. Interactive age and sex distribution visualization from 1950 to 2100.

Title: Japan Population (1950-2025) - Macrotrends
URL: https://www.macrotrends.net/global-metrics/countries/jpn/japan/population
Content: ## Japan Population (1950-2025). Population Economy Trade Health Education Development Labor Force Environment Crime Immigration Other. Population Growth R
ate Density Urban Rural Life Expectancy Birth Rate Death Rate Infant Mortality Rate Fertility Rate. ##### Total current population for Japan in 2025 was **121,960,4
08**, a **0.55% decline** from 2024. * Total population for Japan in 2024 was **122,631,432**, a **1.51% decline** from 2023. * Total population for Japan in 2023 w
as **124,516,650**, a **0.49% decline** from 2022. * Total population for Japan in 2022 was **125,124,989**, a **0.44% decline** from 2021. Total population is base
d on the de facto definition of population, which counts all residents regardless of legal status or citizenship. The values shown are midyear estimates. The full h
istorical dataset is available for download here: Population of Japan | Historical Data | 1950 - 2025.

Title: Japan births hit record low in 2025 | NHK WORLD-JAPAN News
URL: https://www3.nhk.or.jp/nhkworld/en/news/20260226_13/
Content: Japan's health ministry says preliminary figures show that the country recorded a little over 700,000 births last year, a new record low. That is down 15,1
79, or 2.1 percent, from the previous year. The figure marks the tenth consecutive year of decline, and the lowest since record-taking began in 1899. In Tokyo, 88,5
18 babies were born in 2025, an increase of 1,142 from the previous year. In Ishikawa, 6,515 babies were born in 2025, 128 more than in 2024, when the number of bir
ths in the prefecture plunged from the previous year. As for deaths, preliminary figures show that 1,605,654 people died in 2025, down 13,030 from the previous year
. This marks the first decrease in the number of deaths in five years. The ministry says there were 505,656 marriages in 2025, up 5,657 from 2024, marking the secon
d straight year of increase. The number of divorces was 182,969, down 6,983 from the previous year. * [#Japan](/nhkworld/en/news/tags/2/).

Title: As of 2025, Japan has the world's oldest population, with over 25.9 ...
URL: https://www.facebook.com/bloombergopinion/videos/as-of-2025-japan-has-the-worlds-oldest-population-with-over-259-aged-65-or-older/4320040114936800/
Content: # As of 2025, Japan has the world's oldest population, with over 25.9% aged 65 or older. # Video. As of 2025, Japan has the world's oldest population, with
 over 25.9% aged 65 or older. ## **Bloomberg Opinion**. As of 2025, Japan has the world's oldest population, with over 25.9% aged 65 or older. ## Related Reels. Ima
ge 4 Bloomberg Opinion 34K. Image 5 Bloomberg Opinion 7.1K. Image 6 Bloomberg Opinion 2.2K. Image 9 Bloomberg Opinion 1.3M. ## Related Videos. Image 16: Video thumb
```

Comment

- Agent decomposes the query in parts and query for the first entity. Then gets the observations of the query.

```
--- Iteration 2 ---
Thought: I have found multiple sources for Japan's population in 2025. Macrotrends states 121,960,408, CEIC Data states 123.2 million, and PopulationPyramid.net sta
tes 123,103,479. I will use the most precise figure from Macrotrends. Now I need to find Taiwan's population in 2025.
Action: Search["Taiwan population 2025"]
Observation: Title: Taiwan demographics 2026 - StatisticsTimes.com
URL: https://statisticstimes.com/demographics/country/taiwan-demographics.php
Content: As of 2026, Taiwan's population is projected at 23.01 million, compared to 23.11 million in 2025. It ranks 61st in the world by population

Title: Taiwan Population - Trading Economics
URL: https://tradingeconomics.com/taiwan/population
Content: ## The total population in Taiwan was estimated at 23.3 million people in 2025, according to the latest census figures and projections from Trading Economi
cs. Looking back, in the year of 1974, Taiwan had a population of 15.9 million people. ### Population in Taiwan is expected to reach 23.40 Million by the end of 202
6, according to Trading Economics global macro models and analysts expectations. In the long-term, the Taiwan Population is projected to trend around 23.50 Million
in 2027, according to our econometric models. | Unemployment Rate | 3.36 | 3.35 | percent | Jan 2026 |. | Wages | 76495.00 | 62459.00 | TWD/Month | Jan 2026 |. | Wa
ges in Manufacturing | 81740.00 | 66575.00 | TWD/Month | Jan 2026 |. Taiwan Export Orders Hit 6-Month Low. Taiwan Holds Rates as Expected. Taiwan Imports Growth at
Over 1-Year Low. Taiwan Exports Growth Ease to 11-Month Low. Taiwan Trade Surplus at 5-Month Low. Taiwan Inflation Rate Hits 10-Month High.

Title: Taiwan Population (1950-2025) - Macrotrends
URL: https://www.macrotrends.net/global-metrics/countries/twn/taiwan/population
Content: ## Taiwan Population (1950-2025). Population Economy Trade Health Education Development Labor Force Environment Crime Immigration Other. Population Growth
Rate Density Urban Rural Life Expectancy Birth Rate Death Rate Infant Mortality Rate Fertility Rate. ##### Total current population for Taiwan in 2025 was **23,974,
031**, a **0.1% increase** from 2024. * Total population for Taiwan in 2024 was **23,950,214**, a **0.11% increase** from 2023. * Total population for Taiwan in 202
3 was **23,923,276**, a **0.13% increase** from 2022. * Total population for Taiwan in 2022 was **23,893,394**, a **0.14% increase** from 2021. Total population is
based on the de facto definition of population, which counts all residents regardless of legal status or citizenship. The values shown are midyear estimates. The fu
ll historical dataset is available for download here: Population of Taiwan | Historical Data | 1950 - 2025.

Title: Taiwan becomes 'super-aged society' in 2025 - UCA News
URL: https://www.ucanews.com/news/taiwan-becomes-super-aged-society-in-2025/111533
Content: Taiwan has become a "super-aged society" for the first time, with 20 percent of its shrinking population aged 65 or older in 2025, government data showed o
n Jan. 9. The island of more than 23 million people faces a looming demographic challenge, with its population falling overall as well as ageing. There were 4,673,1
55 people aged 65 or older in Taiwan in 2025, or 20.06 percent of the population, according to the interior ministry. The number of people aged 0-14 accounted for 1
1.51 percent of the population, the ministry's data showed. Maharashtra becomes latest Indian state to pass anti-conversion law. advertising, UCA News remains free
— because truth should never come with a price tag. and most trusted Catholic news service, reaching millions worldwide. Your contribution helps keep UCA News a bea
con of truth, faith, and integrity for millions. Maharashtra becomes latest Indian state to pass anti-conversion law. UCA News Catholic Dioceses in Asia. UCA News C
atholic Dioceses in Asia. UCA News Catholic Dioceses in Asia.

Title: Taiwan (China) Population, 1950 – 2026 | CEIC Data
URL: https://www.ceicdata.com/en/indicator/taiwan/population
Content: 1950 - 2025 | Yearly | Person mn | CEIC Data. | 23.3  2025 | 23.4  2024 | 8.0 1950 | 23.6 2019 | Person mn | yearly | 1950 - 2025 |. | China (Person mn) |
1,404.9  2025 | yearly | 1949 - 2025 |. | Taiwan IPI: 1991=100: Mfg: Leather Fur and Products (1991=100) | 30.10  Feb 1999 | monthly | Jan 1982 - Feb 1999 |. | Taiw
```

Comment

- Agent reasons about the results and then request the information of the next entity.

```
00) | 70.45  Feb 1999 | monthly | Jan 1982 - Feb 1999 |. | Taiwan IPI: 1991=100: Mfg: EEM: Data Storage Equipment (1991=100) | 312.72  Jan 1999 | monthly | Jan 1982
 - Jan 1999 |. | Taiwan IPI: 1991=100: Mfg: EEM: Data Storage Media Units (1991=100) | 267.84  Jan 1999 | monthly | Jan 1982 - Jan 1999 |.



--- Iteration 3 ---
Thought: I have found multiple sources for Taiwan's population in 2025. Macrotrends states 23,974,031, Trading Economics states 23.3 million, StatisticsTimes.com st
ates 23.11 million (but projects 23.01 million for 2026), and CEIC Data states 23.3 million. UCA News mentions "over 23 million". I will use the figure from Macrotr
ends as it seems to be the most precise and consistent with other sources mentioning "over 23 million".

Now I have Japan's population as 121,960,408 (from Macrotrends) and Taiwan's population as 23,974,031 (from Macrotrends). I can calculate the fraction.
Final: Taiwan's population in 2025 is approximately 23,974,031 and Japan's population in 2025 is approximately 121,960,408. The fraction of Japan's population that
is Taiwan's population is approximately 23,974,031 / 121,960,408 ≈ 0.1966.
End of execution
```

Comment

- Agent reasons over all the information, notices it can compute the operation. Computes it and delivers final response problems , hard to get data and ambiguous data.

Task 2: Technical Specificity

```
(age_ncu) ghostrader@Guille:~/AGE_NCU/assig2$ python main.py
ReAct Agent Started. Type 'exit' to quit.
Enter your query: Compare the main display specs of iPhone 15 and Samsung S24."

--- Iteration 1 ---
Thought: The user is asking for a comparison of display specs between iPhone 15 and Samsung S24. I need to find the display specifications for both phones and then
compare them. I will start by searching for the display specs of the iPhone 15.
Action: Search["iPhone 15 display specs"]
Observation: Title: Apple iPhone 15 Full Specifications - PhoneArena
URL: https://www.phonearena.com/phones/Apple-iPhone-15_id11927
Content: PhoneArena | Phone News, Reviews, Deals and Specs. # Apple iPhone 15. | Features | Super Retina XDR, HDR support, Oleophobic coating, Scratch-resistant gla
ss, Ambient light sensor, Proximity sensor |. | Learn more on our PhoneArena Display Test Results page. | Learn more on our PhoneArena Battery Test Results page. |
Main camera | 48 MP (Sensor-shift OIS, PDAF) Sensor name: Sony IMX904 Aperture size: F1.6 Focal length: 26 mm Pixel size: 2.0 µm |. | Second camera | 12 MP (Ultra-w
ide) Sensor name: Sony IMX772 Aperture size: F2.4 Focal Length: 13 mm |. | Learn more on our PhoneArena Camera Test Results page. | 3G Bands | 1, 2, 4, 5, 8 |. Disc
over the top rivals of Apple iPhone 15. Your iPhone 16 or older model is now worth less as Apple cuts trade-in values. Samsung's new return strategy is so unusual i
t could tempt even happy Galaxy S26 Ultra users to send their phones back. PhoneArena | Phone News, Reviews, Deals and Specs.

Title: Apple iPhone 15 - Specifications | One New Zealand
URL: https://userguide.one.nz/apple/iphone-15-ios-17/specifications/
Content: # Apple iPhone 15. | Dimensions | 147.6 x 71.6 x 7.8 mm |. ### Screen and keys. | Screen type | XDR OLED touch screen, HDR10, Dolby Vision, 16 million colo
urs |. | Screen size | 6.1 inches |. | Main camera | Dual 48 MP, f/1.6, 26mm (wide) + 12 MP, f/2.4, 13mm, 120˚ (ultra wide) |. | Front camera | Dual 12 MP, f/1.9, 2
3mm (wide) + SL 3D, (depth/biometrics sensor) |. | Video recorder | 4K@24/25/30/60fps, 1080p@25/30/60/120/240fps, HDR, Dolby Vision HDR (up to 60fps), Cinematic mod
e (4K@30fps) |. | Chipset | Apple A16 Bionic (4 nm) |. | CPU | Hexa-core (2x3.46 GHz Everest + 4x2.02 GHz Sawtooth) |. | GPU | Apple GPU (5-core graphics) |. | 3.5
mm jack | No |. ### Connectivity. | Wi-Fi | 802.11 a/b/g/n/ac/6, dual-band |. | Battery type | Non-removable Li-Ion 3349 mAh battery |. Excellent Very Good Good Fai
r Low No Connection. Excellent Very Good Good Fair Low No Connection.

Title: Apple iPhone 15 - Specifications | Vodafone UK
URL: https://deviceguides.vodafone.co.uk/apple/iphone-15-ios-17/specifications/
Content: | Dimensions | 147.6 x 71.6 x 7.8 mm |. | Screen type | XDR OLED touch screen, HDR10, Dolby Vision, 16 million colours |. | Resolution | 1179 x 2556 pixels
 |. | Main camera | Dual 48 MP, f/1.6, 26mm (wide) + 12 MP, f/2.4, 13mm, 120˚ (ultra wide) |. | Front camera | Dual 12 MP, f/1.9, 23mm (wide) + SL 3D, (depth/biomet
rics sensor) |. | Video recorder | 4K@24/25/30/60fps, 1080p@25/30/60/120/240fps, HDR, Dolby Vision HDR (up to 60fps), Cinematic mode (4K@30fps) |. | SIM type | Nano
 SIM; eSIM |. | GPU | Apple GPU (5-core graphics) |. | Battery type | Non-removable Li-Ion 3349 mAh battery |. ## Help and support. ## Vodafone UK. Registered Offic
e: Vodafone House, The Connection, Newbury, Berkshire, RG14 2FN. \*Annual Price Increase: During your minimum period, the monthly cost will increase each year on 1
April by £1.80 for Pay monthly plans with Airtime/Data, and £3 for Home Broadband plans. More information: vodafone.co.uk/pricechanges.

Title: Apple iPhone 15 full review - YouTube
URL: https://www.youtube.com/watch?v=T8ZnoMQpimA
Content: Beyond the move to a USB-C port, the iPhone 15 has ditched the notch and comes ... - Intro 00:33 - Design and build 01:54 - Screen specs and
```

Comment

- Agent understand task and decompose it, then request action. Proceeds to get the observation.

```
--- Iteration 2 ---
Thought: I have gathered display specifications for the iPhone 15. Now I need to find the display specifications for the Samsung S24.
Action: Search["Samsung S24 display specs"]
Observation: Title: Samsung S24 series tech specs (S24, S24+, S24 FE, S24 Ultra)
URL: https://www.backmarket.com/en-us/c/samsung/samsung-s24-tech-specs?srsltid=AfmBOooEZgR9zcC_gTT7bf3GxNNugNl8pBWxAudUpW24NwQ3GBdoDY9-
Content: # Samsung S24 series tech specs (S24, S24+, S24 FE, S24 Ultra). We're here to give you more information on the Samsung S24 series release date, camera spec
s, and overall features. Under the hood, the Galaxy S24+ is powered by a Qualcomm Snapdragon 8 Gen 3 processor, and other specs include a triple camera setup and a
4,900mAh battery. ## Samsung Galaxy S24 FE specs. The Samsung Galaxy S24 FE introduces a 6.7-inch Dynamic AMOLED display with a smooth 120Hz refresh rate, with the
Exynos 2400e processor under the hood. ## Samsung Galaxy S24 Ultra specs. The Samsung Galaxy S24 Ultra sits at the top of the pyramid with a 6.8-inch Dynamic AMOLED
 display with a 120Hz refresh rate. The Samsung S24 Ultra features a 5,000mAh battery, and its camera specs include a quad camera setup on the back. + **Video captu
re:** 3840x2160 (4K UHD) (60 fps). You can purchase a Samsung S24, S24+, S24 FE, or S24 Ultra new or refurbished.

Title: Samsung Galaxy S24 Full Specifications - PhoneArena
URL: https://www.phonearena.com/phones/Samsung-Galaxy-S24_id12113
Content: Size, 6.2-inch, 90.87% screen-to-body ; Type, Dynamic AMOLED, 120Hz, HDR, 2600 nits ; Resolution, 2340x1080px, 19.5:9 ratio, 416 PPI ; Protection

Title: Galaxy S24: viewport, screen size, CSS pixel ratio, cross ... - Blisk
URL: https://blisk.io/devices/details/galaxy-s24
Content: Galaxy S24 has a 6.2-inch screen with a screen size (resolution): 1080px × 2340px , 360px × 780px viewport1, and a CSS Pixel Ratio of 3. **Device Pixel Rat
io (DPR)** or **CSS Pixel Ratio** is the ratio between the physical pixels (screen size or resolution) and CSS pixels (viewport). Screen Size (Resolution) = Viewpor
t size × Device Pixel Ratio. Viewport size = Screen Size (Resolution) / Device Pixel Ratio. You also need to consider that the users will manipulate your web applic
ation on Galaxy S24 with a finger and stylus because this device has a touch screen that supports touch events. Galaxy S24 is ready to use in the Blisk app, where y
ou can develop web applications and test cross-browser compatibility, use this phone as a standalone device or simultaneously with other devices. You can launch Gal
axy S24 in the Blisk app from Command prompt (Windows) or Terminal (macOS or Linux) to develop web applications and test cross-browser compatibility on a standalone
 device or simultaneously with other devices.

Title: Samsung Galaxy S24 (Exynos) Display test - DXOMARK
URL: https://www.dxomark.com/samsung-galaxy-s24-display-test/
Content: # Samsung Galaxy S24 (Exynos) Display test. We put the Samsung Galaxy S24 through our rigorous DXOMARK Display test suite to measure its performance across
 six criteria. The Samsung Galaxy S24 nearly matched the display performance of the top-of-the-line S24 Ultra, thanks to achieving two top scores in readability and
 touch, but the smaller device was held back due to its management of artifacts. ***About DXOMARK Display tests**: For scoring and analysis in our smartphone and ot
her display reviews, DXOMARK engineers perform a variety of objective and perceptual tests under controlled lab and real-life conditions.* *Note that we evaluate di
splay attributes using only the device's built-in display hardware and its still image (gallery) and video apps at their default settings.* *(For in-depth informati
on about how we evaluate smartphone and other displays, check out our articles, "How DXOMARK tests display quality" and "A closer look at DXOMARK Display testing."*
. Samsung Galaxy S24, color fidelity at 1000 lux in the Display-P3 color space.

Title: Samsung Galaxy S24 review - YouTube
URL: https://www.youtube.com/watch?v=u9yv3RmqAm8
```

Comment

- Reflects on the observation and notices need information of others parts. Request action and gets information.

```
--- Iteration 3 ---
Thought: I have gathered display specifications for both the iPhone 15 and the Samsung S24. Now I need to extract and compare the key display features.

iPhone 15 Display Specs:
- Type: Super Retina XDR OLED, HDR10, Dolby Vision
- Size: 6.1 inches
- Resolution: 1179 x 2556 pixels
- Refresh Rate: Not explicitly stated in the provided snippets, but typically 60Hz for the base model.
- Brightness: Not explicitly stated, but Super Retina XDR implies high brightness.

Samsung S24 Display Specs:
- Type: Dynamic AMOLED, 120Hz, HDR
- Size: 6.2 inches
- Resolution: 2340 x 1080 pixels
- Refresh Rate: 120Hz
- Brightness: 2600 nits (peak)

Now I will compare these specs.
Action: Search["iPhone 15 display refresh rate and brightness"]
Observation: Title: Iphone 15 display herz rate - Apple Support Community
URL: https://discussions.apple.com/thread/255626726
Content: The iPhone 15 does not have ProMotion feature and therefore does not have an adaptive refresh rate. The iPhone 15 operates on a 60Hz refresh rate.

Title: Do you find the difference in quality between the iPhone 15's 60hz ...
URL: https://www.quora.com/Do-you-find-the-difference-in-quality-between-the-iPhone-15s-60hz-and-the-15-Pro-Maxs-120hz-screen-significant
Content: No, iPhone 15 and 15 Plus doesent support 120Hz display. But the iPhone 15 pro and pro Max do have a 120 Hz display refresh rate. Continue

Title: iPhone 15 refresh rate - Reddit
URL: https://www.reddit.com/r/iphone/comments/172szy1/iphone_15_refresh_rate/
Content: In your opinion, is the 60hz refresh rate more noticeable with a large or small display? iPhone 15 or 15 Plus? Thanks in advance.

Title: Capping iPhone 15 Pro/Pro Max refresh rate at 60Hz ... - PhoneArena
URL: https://www.phonearena.com/news/reducing-iphone-15-pro-refresh-rate-60hz-adds-limited-amount-battery-life_id152661
Content: # Capping iPhone 15 Pro/Pro Max refresh rate at 60Hz extends battery life by a small percentage. Lowering the refresh rate maximum to 60Hz on a compatible
iPhone Pro model might give you 5% or 6% longer battery life - Capping iPhone 15 Pro/Pro Max refresh rate at 60Hz extends battery life by a small percentage. ## Is
it worth the extra battery life to disable the 120Hz refresh rate on iPhone 15 Pro models? Lowering the refresh rate maximum to 60Hz on a compatible iPhone Pro mode
l might give you 5% or 6% longer battery life - Capping iPhone 15 Pro/Pro Max refresh rate at 60Hz extends battery life by a small percentageLowering the refresh ra
te maximum to 60Hz on a compatible iPhone Pro model might give you 5% or 6% longer battery life - Capping iPhone 15 Pro/Pro Max refresh rate at 60Hz extends battery
 life by a small percentage. Capping iPhone 15 Pro/Pro Max refresh rate at 60Hz extends battery life by a small percentage.
```

Comment

- Now with the complete information, it starts comparing it. Notices it lacks some information, request further Action and get observation back.

```
--- Iteration 4 ---
Thought: I have confirmed that the iPhone 15 has a 60Hz refresh rate. The previous search for Samsung S24 display specs mentioned a 120Hz refresh rate and 2600 nits
 brightness. I now have enough information to compare the main display specs.

iPhone 15 Display Specs:
- Type: Super Retina XDR OLED, HDR10, Dolby Vision
- Size: 6.1 inches
- Resolution: 1179 x 2556 pixels
- Refresh Rate: 60Hz
- Brightness: Not explicitly stated, but Super Retina XDR implies high brightness.

Samsung S24 Display Specs:
- Type: Dynamic AMOLED, HDR
- Size: 6.2 inches
- Resolution: 2340 x 1080 pixels
- Refresh Rate: 120Hz
- Brightness: 2600 nits (peak)

I can now present the comparison.
Final: The main display specifications for the iPhone 15 and Samsung S24 are as follows:

**iPhone 15:**
*   **Type:** Super Retina XDR OLED, HDR10, Dolby Vision
*   **Size:** 6.1 inches
*   **Resolution:** 1179 x 2556 pixels
*   **Refresh Rate:** 60Hz
*   **Brightness:** Not explicitly stated, but Super Retina XDR implies high brightness.

**Samsung S24:**
*   **Type:** Dynamic AMOLED, HDR
*   **Size:** 6.2 inches
*   **Resolution:** 2340 x 1080 pixels
*   **Refresh Rate:** 120Hz
*   **Brightness:** 2600 nits (peak)

Key differences include the refresh rate (60Hz on iPhone 15 vs. 120Hz on S24), screen size (6.1" vs. 6.2"), and peak brightness (S24 has a significantly higher stat
ed peak brightness of 2600 nits). The iPhone 15 has a slightly higher resolution.
End of execution
```

Comment

- Reason on the new information, since has all the elements, now it can finalize the answer.

Task 3: Resilience & Reflection Test

```
(age_ncu) ghostrader@Guille:~/AGE_NCU/assig2$ python main.py
ReAct Agent Started. Type 'exit' to quit.
Enter your query: Who is the CEO of the startup 'Morphic' AI search?

--- Iteration 1 ---
Thought: The user is asking for the CEO of a specific startup. I need to find this information. I will search for "Morphic AI search CEO".
Action: Search["Morphic AI search CEO"]
Observation: Title: The SaaS Pulse Show - Episode #1 - Weston Baker - Founder-CEO
URL: https://www.youtube.com/watch?v=WJxH8Ru9JKs
Content: Weston Baker, Founder-CEO, Morphic https://www.linkedin ... How Open Source, Python and AI Are Shaping the Data Future with Wes McKinney.

Title: Jaynti Kanani (JD) (@jdkanani) / Posts / X
URL: https://x.com/jdkanani
Content: Jaynti Kanani (JD) (@jdkanani) - Posts - Founder & CEO at @morphic . Previously, Founder & CEO @0xPolygon | X (formerly Twitter)

Title: Morphic - 2026 Company Profile & Team - Tracxn
URL: https://tracxn.com/d/companies/morphic/__QREWaDzuHM-XW-PFwSgyqKupVzTD3aOF5VD9L0I9mSw
Content: Your browser was unable to load all of Tracxn resources. # Morphic - Company Profile. ## Morphic - About the company. Morphic is an unfunded company based
in San Francisco (United States), foundedbyJaynti Kanani. Morphic has not raised any funding yet. View Morphic's full profile. Sign up to download Morphic's company
 profile. ## Morphic's funding and investors. Morphic has not raised any funding rounds yet. ## Morphic's founders and board of directors. * Jaynti Kanani: Founder
& CEO of Morphic and founder of 1 other company. ## Morphic's Investments and acquisitions. ## Reports related to Morphic. Viewall reports related to Morphic. View
complete company profile of Morphic. ## FAQs about Morphic. * GAYA BAZAAR - Gaya based, 2023 founded, Unfunded company. * Nexogenius Technologies - Rajkot based, 20
23 founded, Unfunded company. * Xcelsior Data Services - Puducherry based, 2022 founded, Unfunded company. * Ptc Shop - Australia based, 2009 founded, Unfunded comp
any. * TCI Chemicals - Tokyo based, 1946 founded, Unfunded company.

Title: Weston Baker - Founder and CEO @ Morphic | Whitebridge.AI
URL: https://whitebridge.ai/contacts/weston-baker-email-332093?srsltid=AfmBOoqgmSKN6if5lINavxfly7vPuu_v5UlyplSi2Fxw9NFzXCvS2Hzz
Content: ## Weston Baker Work. Information regarding Weston Baker's phone number may be available. ### Where is Weston Baker located? ## Weston Baker Education. ##
Weston Baker Skills. ## Frequently Asked Questions about Weston Baker. ### What is Weston Baker's email address? # Weston Baker. View Weston Baker's Email & Phone N
umber;). We have information regarding potential email addresses for Weston Baker. ### How to contact Weston Baker? You can try to contact Weston Baker via email or
 phone. ### What company does Weston Baker work for? Weston Baker's current employer information is not specified in detail here. ### What industry does Weston Bake
r work in? Weston Baker's primary industry can often be inferred from their work experience and skills listed on this profile. ### What is Weston Baker's phone numb
er? Weston Baker is based in or around New York, NY, US. ## Weston Baker Email Addresses:. ## Weston Baker Phone Numbers:. ## People like Weston Baker. ## Not the W
eston Baker you were looking for?

Title: "Harness those cross-functional skills" with Weston Baker of Morphic
URL: https://www.youtube.com/watch?v=daorSuHJNpg
Content: Weston Baker, Founder & CEO of Morphic and an overall force in the ... ai/blog/what-the-tech-episode-21-harness-cross-functional-skills
```

Comment

- Agent is awaken, user asks a question, agent generates thought, then an action and finishes with an observation of the query.

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Title: "Harness those cross-functional skills" with Weston Baker of Morphic
URL: https://www.youtube.com/watch?v=daorSuHJNpg
Content: Weston Baker, Founder & CEO of Morphic and an overall force in the ... ai/blog/what-the-tech-episode-21-harness-cross-functional-skills



--- Iteration 2 ---
Thought: The search results provide conflicting information. One result states Jaynti Kanani is the Founder & CEO of Morphic, while others mention Weston Baker as F
ounder-CEO of Morphic. I need to clarify which one is correct or if there are multiple people with similar roles or if one is more current. The Tracxn result seems
to be a company profile and lists Jaynti Kanani as Founder & CEO. The YouTube links and Whitebridge.AI link mention Weston Baker as Founder & CEO. I will prioritize
 the Tracxn company profile as it's likely to be more structured and verified company data. However, to be absolutely sure, I will search for "Jaynti Kanani Morphic
" and "Weston Baker Morphic" to see if there's any further clarification or if one is more recent.
Action: Search["Jaynti Kanani Morphic CEO"]
Observation: Title: Jaynti Kanani - Founder & CEO at Morphic - The Org
URL: https://theorg.com/org/morphic-1/org-chart/jaynti-kanani
Content: Currently, Jaynti holds the position of Founder & CEO at both Mozak and Morphic, continuing to make significant strides in the field. Kanani's educational

Title: Jaynti Kanani - Contra
URL: https://contra.com/jd000007/about
Content: Founder & CEO at Morphic. https://morphic.com/. Dubai - United Arab Emirates. Transforming the future of storytelling using breakthrough technologies.

Title: JD Kanani is our newest judge for the Meta Morph Award 2025.   As ...
URL: https://www.instagram.com/p/DHvZYZwvu_S/
Content: As the Founder/CEO of Morphic, a developing platform designed to empower creators, filmmakers, and animators to produce high-quality stories

Title: Jaynti Kanani (JD) (@jdkanani) / Posts / X
URL: https://x.com/jdkanani
Content: Jaynti Kanani (JD) (@jdkanani) - Posts - Founder & CEO at @morphic . Previously, Founder & CEO @0xPolygon | X (formerly Twitter)

Title: Polygon Cofounder Jaynti Kanani Steps Down To Pursue New ...
URL: https://inc42.com/buzz/polygon-cofounder-jaynti-kanani-steps-down-to-focus-on-new-adventures/
Content: As per Kanani's LinkedIn profile, he served as the cofounder of Polygon till March 2023 and is now the founder and CEO of presumably two new startups – Mor
phic
```

Comment

- Agent reflects on the observation, its not clear, adjust it search for a more narrow action. This excels a new observation.

```
--- Iteration 3 ---
Thought: The search results consistently identify Jaynti Kanani as the Founder & CEO of Morphic. The previous results mentioning Weston Baker might have been outdat
ed or referred to a different entity. The latest information points to Jaynti Kanani.
Final: Jaynti Kanani is the CEO of the startup 'Morphic' AI search.
End of execution
```

Comment

- Based on consistent results, agent decides the information is enough to finish the search.

*Screen shots dont show the full content provided in each observation to safe space, the content is the information provided from the Tavily API, so having every single line of it seemed inconsequential.